

2m Temperature (land-only) [celsius]	13.50 13.70	2.61 2.65	23.71 24.24	14.70 14.66	5.37 6.53	-12.66 -11.46	21.34 22.70	21.43 20.51	20.88 20.19	17.76 16.59	24.71 24.60	7.77 8.66
Mean Sea Level Pressure [hPa]	1011.14 1011.43	1016.27 1014.76	1012.01 1013.20	1004.19 1004.66	1011.46 1011.58	1017.87 1015.89	1011.36 1012.99	1005.24 1004.55	1010.83 1011.30	1013.74 1012.86	1013.07 1013.66	1003.22 1005.09
Precipitation [mm/day]	2.70 2.85	1.92 2.15	3.38 3.40	2.08 2.45	2.69 2.80	2.02 2.27	3.48 3.44	1.69 2.06	2.78 2.93	1.88 2.12	3.36 3.40	2.46 2.80
Evaporation [mm/day]	-2.71 -2.91	-1.47 -1.74	-3.78 -3.95	-1.71 -1.99	-2.70 -2.87	-1.52 -1.76	-3.87 -3.99	-1.43 -1.74	-2.79 -3.01	-1.46 -1.76	-3.84 -4.03	-1.95 -2.20
Precip. minus Evap. [mm/day]	-0.01 0.02	0.45 0.53	-0.40 -0.47	0.37 0.50	-0.01 0.04	0.50 0.61	-0.38 -0.45	0.26 0.42	-0.02 -0.01	0.42 0.51	-0.48 -0.56	0.51 0.57
Total Cloud Cover [frac]	0.65 0.68	0.70 0.68	0.57 0.62	0.74 0.79	0.65 0.67	0.75 0.69	0.58 0.62	0.71 0.77	0.64 0.67	0.66 0.66	0.57 0.62	0.78 0.79
Low Cloud Cover [frac]	0.34 0.38	0.43 0.44	0.22 0.27	0.49 0.56	0.35 0.39	0.46 0.49	0.22 0.26	0.50 0.53	0.34 0.38	0.44 0.40	0.23 0.27	0.48 0.57
Medium Cloud Cover [frac]	0.31 0.25	0.41 0.32	0.19 0.16	0.48 0.35	0.33 0.26	0.49 0.38	0.20 0.17	0.42 0.31	0.30 0.24	0.30 0.26	0.20 0.17	0.52 0.38
High Cloud Cover [frac]	0.41 0.34	0.42 0.34	0.39 0.34	0.44 0.34	0.41 0.34	0.45 0.34	0.41 0.35	0.38 0.31	0.41 0.34	0.37 0.33	0.38 0.33	0.51 0.36
Precipitation (ocean) [Sv]	11.84 13.39	1.70 2.10	7.28 7.86	2.86 3.44	11.89 13.31	2.20 2.56	7.40 7.88	2.29 2.88	12.12 13.48	1.18 1.67	7.52 7.88	3.42 3.93
Precip. minus Evap. (ocean) [Sv]	-1.88 -1.28	0.25 0.33	-2.58 -2.25	0.46 0.62	-2.20 -1.46	0.16 0.18	-2.69 -2.24	0.33 0.57	-1.53 -1.22	0.46 0.66	-2.63 -2.58	0.64 0.68
Precipitation (land) [Sv]	4.11 3.44	1.07 1.07	2.89 2.19	0.15 0.19	3.97 3.25	0.72 0.80	3.10 2.29	0.15 0.16	4.27 3.81	1.53 1.45	2.61 2.16	0.13 0.20
Precip. minus Evap. (land) [Sv]	1.84 1.41	0.40 0.46	1.37 0.84	0.07 0.11	2.15 1.70	0.57 0.73	1.53 0.92	0.04 0.05	1.43 1.18	0.15 0.09	1.19 0.93	0.09 0.17
TOA Net [W/m2]	6.23 0.90	-38.12 -43.38	46.07 45.03	-32.59 -43.10	13.60 7.92	-119.88 -128.23	50.83 50.40	69.36 59.10	-3.22 -7.50	49.30 46.85	28.54 27.81	-122.01 -132.48
TOA SW Net [W/m2]	239.11 241.34	175.29 180.65	302.06 305.53	171.52 173.64	243.99 245.58	72.31 74.74	304.79 309.38	288.74 288.82	232.89 236.51	286.99 293.85	285.42 289.48	69.12 73.23
TOA LW Net [W/m2]	-232.88 -240.44	-213.41 -224.03	-255.99 -260.50	-204.11 -216.75	-230.39 -237.66	-192.19 -202.97	-253.96 -258.98	-219.38 -229.73	-236.10 -244.01	-237.69 -247.00	-256.88 -261.67	-191.13 -205.71
TOA SW Net (clear sky) [W/m2]	285.33 287.05	218.35 224.07	346.61 348.53	224.39 227.04	293.55 295.12	90.72 95.21	351.86 353.90	374.64 377.49	278.83 280.40	360.18 363.27	329.26 331.56	92.20 95.19
TOA LW Net (clear sky) [W/m2]	-258.99 -268.35	-238.25 -248.53	-281.89 -290.60	-231.90 -243.67	-255.82 -265.44	-216.12 -226.83	-280.82 -290.10	-243.33 -254.74	-262.73 -271.79	-262.21 -271.79	-282.24 -290.62	-222.51 -234.14
SW Cloud Forcing [W/m2]	-46.22 -45.70	-43.06 -43.42	-44.56 -43.00	-52.87 -53.40	-49.56 -49.54	-18.41 -20.46	-47.07 -44.52	-85.90 -88.67	-45.94 -43.88	-73.19 -69.42	-43.84 -42.08	-23.08 -21.95
LW Cloud Forcing [W/m2]	26.11 27.91	24.84 24.50	25.90 30.10	27.79 26.92	25.43 27.78	23.93 23.86	26.86 31.12	23.95 25.01	26.62 27.78	24.51 24.79	25.36 28.95	31.38 28.43
Surface Net SW [W/m2]	161.01 160.00	110.56	208.36	112.59	165.87	44.92	211.27	192.03	154.14	180.62	194.30	43.83
Surface Net LW [W/m2]	-58.40 -56.00	-54.80	-63.73	-50.89	-57.67	-52.68	-63.69	-50.09	-58.01	-53.60	-63.12	-51.77
Surface SH [W/m2]	-17.56 -21.00	-19.63	-20.39	-9.57	-16.42	-18.67	-19.89	-6.94	-19.18	-22.47	-20.73	-12.66
Surface LH [W/m2]	-78.50 -82.00	-42.80	-109.41	-49.67	-78.11	-43.98	-112.01	-41.44	-80.88	-42.40	-111.05	-56.39
Net Surface (no snowfall) [W/m2]	6.55 0.60	-6.67	14.84	2.46	13.67	-70.41	15.67	93.55	-3.94	62.14	-0.60	-76.99
Net Surface [W/m2]	5.79 0.60	-8.06	14.83	0.77	12.81	-72.97	15.65	92.66	-4.56	61.90	-0.60	-79.28
TOA - Sfc Net Radiation (no snow) [W/m2]	-0.32 0.00	-31.45	31.23	-35.05	-0.07	-49.47	35.15	-24.19	0.72	-12.84	29.13	-45.03
TOA - Sfc Net Radiation [W/m2]	0.44 0.00	-30.06	31.24	-33.36	0.79	-46.91	35.18	-23.30	1.34	-12.61	29.13	-42.73
	ALL Global	ALL North Midlat	ALL Tropical	ALL South Midlat	DJF Global	DJF North Midlat	DJF Tropical	DJF South Midlat	JJA Global	JJA North Midlat	JJA Tropical	JJA South Midlat

Model Bias
(standard deviation of interannual variability from observations)

10
9
8
7
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
-7
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-9
-10